



KUALA LUMPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Assignment 1: Case Analysis Report

Ethical & Legal Analysis of an IT-Related Case: British Airways Data Breach

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Declaration of AI Usage

This assignment includes content assisted by AI (Gemini) for outlining the structure, simplifying legal concepts. The final writing and analysis are my own work.

1. Introduction

Data protection is very important in the modern world. When companies do not protect their systems, they put their customers at risk. This report analyzes the British Airways (BA) data breach that happened in 2018. It was a serious cyber-attack where hackers stole the personal and financial details of hundreds of thousands of customers. The objective of this report is to summarize the case, identify the relevant laws (like the GDPR), analyze the ethical issues, and give recommendations to stop this from happening again.

2. Main Content

i. Summary of the Case

The British Airways data breach happened between August 21, 2018, and September 5, 2018. The attack was carried out by a hacker group called "Magecart" (Huntress, 2025). The hackers did not break into the main database directly. Instead, they used a "supply chain attack." They stole the login details of a third-party user because the account did not have Multi-Factor Authentication (Information Commissioner's Office, 2020).

Once they were inside the system, the hackers put a malicious script (bad code) on the British Airways website. This script worked like a digital trap. When customers typed their credit card details to buy a ticket, the script sent that data to the hackers (BBC News, 2018). The breach affected about 400,000 customers. The stolen data included names, addresses, emails, and credit card numbers, including the "CVV" security code on the back (The Guardian, 2018).

ii. Relevant Laws

The most important law in this case is the **General Data Protection Regulation (GDPR)**. This law protects the data of people in the UK and Europe.

- **Article 32 of GDPR:** This rule says that companies must use appropriate technical measures to keep data safe.
- **The Extraterritorial Scope of the GDPR:** The legal component of this matter also addresses how far-reaching the GDPR is. It has jurisdiction over all organizations that process information about EU citizens, regardless of their location in the world. Therefore, since British Airways is an international airline; they were compelled to operate according to the

very stringent standards to protect all of their passengers. The case is also a cautionary tale for international companies, that cannot disregard the privacy laws in place in each region, because if an organization does not do so, it will be penalized according to its global revenue, which is calculated by a percentage of its total annual turnover.

- **UK Data Protection Act 2018:** This is the UK version of the GDPR.

British Airways broke these laws because their security was poor. The investigation by the Information Commissioner's Office (ICO) found that BA stored some passwords in "plain text" files without encryption (Clifford Chance, 2020). This is against the law because it makes data easy to steal. Because of this negligence, the ICO fined British Airways £20 million (Information Commissioner's Office, 2020).

iii. Ethical Issues

We can analyze this case using ethical theories to see if British Airways acted correctly.

- **Duty of Care:**

Companies have a "duty of care" to protect their customers. When a passenger buys a ticket, they trust the airline with their money and identity. British Airways failed this duty. They did not use basic security tools like Multi-Factor Authentication (MFA). Ethically, they were negligent because they did not try hard enough to protect the user data (Huntress, 2025).

- **Utilitarianism:**

This theory looks at the "greater good." We compare the good results and the bad results. In this case, BA might have saved money by not upgrading their security. However, the result was that 400,000 people suffered stress and financial risk (BBC News, 2018). The harm to the customers was much bigger than the benefit to the company. Therefore, the lack of security was unethical.

- **Deontology (Duty-Based Ethics)** Actions are evaluated based on whether they comply with certain rules/duties, regardless of their outcomes from a Deontological viewpoint. Within the ACM Code of Ethics, there are two key moral obligations that exist for IT professionals: "Respect Privacy" and "Honor Confidentiality". British Airways also had a very clear ethical obligation to implement the most advanced security practices available. British Airways' failure to encrypt passwords and utilize multi-factor authentication (MFA) violated a basic "duty" to its passengers, which in deontology, could never be justified as an excuse to save money at the expense of other people's safety.

iv. Stakeholder Analysis.

The breach affected different types of stakeholders as follows;

- **The customer:** Over 400,000 customers had the immediate potential for financial fraud. They also experienced "Identity Stress" due to the fact that criminals now have their private address and security code information.
- **The Company (British Airways):** BA has suffered significantly with respect to reputation loss. While the £20million was lost to the fine, many additional millions were spent by them to cover legal fees and repair their compromised IT system.
- **The IT Industry:** This incident was an example of what can happen when companies do not take Cyber Security seriously enough and will likely cause other airlines to assess Cyber Security threats to their own supply chain.

v. Evaluation and Recommendations

Evaluation

In my opinion, this data breach was avoidable. The hackers were inside the system for two weeks before anyone noticed. This shows that the security monitoring at British Airways was very weak. It is also shocking that a big company stored passwords in plain text files. This is a basic mistake that should not happen in a professional IT environment.

Recommendations

To prevent this in the future, I recommend the following:

1. **Use Multi-Factor Authentication (MFA):** All employees and third-party suppliers must use MFA to log in. This stops hackers from using stolen passwords.
2. **Regular Code Audits:** The IT team must check the website code regularly. They should use tools to find unauthorized scripts like the one Magecart used.
3. **Encryption:** Never store passwords or sensitive data in plain text. All data must be encrypted so hackers cannot read it.

3. Conclusion

In conclusion, the British Airways data breach is a big lesson for all companies. It shows that bad security can lead to huge fines and a loss of trust. British Airways failed to follow the GDPR and failed ethically to protect their customers. The £20 million fine sends a clear message that data protection is a serious legal and ethical requirement (Information Commissioner's Office, 2020).

4. References

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